

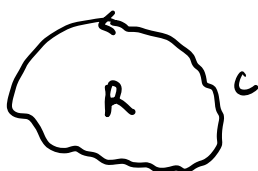
Lecture 22 Stokes' Theorem

Recap: Surface flux

$$\vec{F} = \langle F_x, F_y, F_z \rangle$$

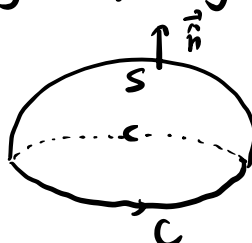
Surface $S: \vec{r}(u, v)$ or $z = g(x, y)$ on region R .

$$\begin{aligned} \iint_S \vec{F} \cdot \vec{n} \, ds &= \pm \iint_R \vec{F} \cdot \left(\frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} \right) du \, dv \\ &= \pm \iint_R \left(-F_x \frac{\partial z}{\partial x} - F_y \frac{\partial z}{\partial y} + F_z \right) dx \, dy. \end{aligned}$$



Green's Theorem

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_R (\nabla \times \vec{F}) \cdot \vec{k} \, dA$$



Stoke's Theorem

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot \vec{n} \, ds$$

Theorem (Stokes' Theorem) Let S be an orientable, simply-connected surface in $\mathbb{R}^3$, and C be the boundary curve of the surface S . Suppose $\vec{F}$ is a vector field defined on C and on the surface S , then we have:

$$\underbrace{\oint_C \vec{F} \cdot d\vec{r}}_{\text{line integral}} = \underbrace{\iint_S (\nabla \times \vec{F}) \cdot \vec{n} \, ds}_{\text{surface integral}},$$

where $\vec{n}$ is the unit normal vector to S , with direction determined by the right-hand grab rule.

Example. Let S be the hemisphere $x^2 + y^2 + z^2 = 4$, $z \geq 0$ above the xy -plane, and C be its boundary curve oriented counter-clockwise on the xy -plane.

Given $\vec{F} = (z - y)\vec{i} + x\vec{j} - x\vec{k}$. Evaluate $\oint_C \vec{F} \cdot d\vec{r}$.

Sol. $\oint \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot \vec{n} \, ds$

$$\vec{r}(\theta, \phi) = (2 \sin \phi \cos \theta) \vec{i} + (2 \sin \phi \sin \theta) \vec{j} + 2 \cos \phi \vec{k}.$$

$$0 \leq \theta < 2\pi, \quad 0 \leq \phi \leq \frac{\pi}{2}.$$

$$\frac{\partial \vec{r}}{\partial \theta} = (-2 \sin \phi \sin \theta) \vec{i} + (2 \sin \phi \cos \theta) \vec{j} + 0 \vec{k}$$

$$\frac{\partial \vec{r}}{\partial \phi} = (2 \cos \phi \cos \theta) \vec{i} + (2 \cos \phi \sin \theta) \vec{j} - 2 \sin \phi \vec{k}.$$

$$\frac{\partial \vec{r}}{\partial \theta} \times \frac{\partial \vec{r}}{\partial \phi} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -2 \sin \phi \sin \theta & 2 \sin \phi \cos \theta & 0 \\ 2 \cos \phi \cos \theta & 2 \cos \phi \sin \theta & -2 \sin \phi \end{vmatrix}$$

$$= (-4 \sin^2 \phi \cos \theta) \vec{i} + (-4 \sin^2 \phi \sin \theta) \vec{j} + (-4 \sin \phi \cos \phi) \vec{k}.$$

$$\nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ z-y & x & -x \end{vmatrix} = 2 \vec{j} + 2 \vec{k}.$$

Using the Stokes' Theorem

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot \vec{n} \, ds$$

$$= \int_0^{\pi/2} \int_0^{2\pi} (2 \vec{j} + 2 \vec{k}) \cdot \left(-\frac{\partial \vec{r}}{\partial \theta} \times \frac{\partial \vec{r}}{\partial \phi} \right) d\theta d\phi$$

$$= \int_0^{\pi/2} \int_0^{2\pi} (8 \sin^2 \phi \sin \theta + 8 \sin \phi \cos \phi) d\theta d\phi.$$

$$= 8 \left(\int_0^{\pi/2} \sin^2 \phi d\phi \right) \underbrace{\left(\int_0^{2\pi} \sin \theta d\theta \right)}_0 + 2\pi \int_0^{\pi/2} 4 \sin 2\phi d\phi$$

$$= 8\pi \left[-\frac{1}{2} \cos 2\phi \right]_0^{\pi/2}$$

$$= 8\pi.$$

Direct computation.

$$\vec{r}(t) = (2 \cos t) \vec{i} + (2 \sin t) \vec{j} \quad 0 \leq t \leq 2\pi.$$

$$\begin{aligned}
 \oint_C \vec{F} \cdot d\vec{r} &= \oint_C \vec{F} \cdot \vec{r}'(t) dt \\
 &= \int_0^{2\pi} ((z-y)\vec{i} + x\vec{j} - x\vec{k}) \cdot ((-2\sin t)\vec{i} + (2\cos t)\vec{j} + 0\vec{k}) dt \\
 &= \int_0^{2\pi} (4\sin t + 4\cos t) dt \\
 &= 8\pi
 \end{aligned}$$

Proof of Stokes' Theorem.

Special case: S is the graph of the function $z = g(x, y)$

$$\vec{F} = \langle F_x, F_y, F_z \rangle.$$

$$\oint_C \vec{F} \cdot d\vec{r} = \oint F_x dx + F_y dy + F_z dz. \quad dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy$$

$$= \oint_C F_x dx + F_y dy + F_z \left(\frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy \right)$$

$$= \oint_C \underbrace{(F_x + F_z \frac{\partial z}{\partial x})}_{M(x,y)} dx + \underbrace{(F_y + F_z \frac{\partial z}{\partial y})}_{N(x,y)} dy.$$

$$\text{Let } \vec{u}(x, y) = \langle M(x, y), N(x, y) \rangle.$$

$$\text{By Green's Theorem, } \oint_C M dx + N dy$$

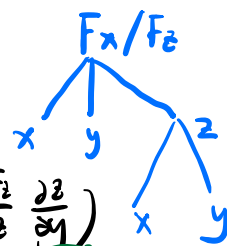
$$= \oint_C \vec{u} \cdot d\vec{r}$$

$$= \iint_R (\nabla \times \vec{u}) \cdot \vec{k} dA$$

$$= \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dA.$$

$$\frac{\partial M}{\partial y} = \frac{\partial F_x}{\partial y} + \frac{\partial F_x}{\partial z} \frac{\partial z}{\partial y} + \underbrace{F_z \cdot \frac{\partial^2 z}{\partial y \partial x}} + \frac{\partial z}{\partial x} \left(\frac{\partial F_z}{\partial y} + \frac{\partial F_z}{\partial z} \frac{\partial z}{\partial y} \right)$$

$$\frac{\partial N}{\partial x} = \frac{\partial F_y}{\partial x} + \frac{\partial F_y}{\partial z} \frac{\partial z}{\partial x} + \underbrace{F_z \frac{\partial^2 z}{\partial x \partial y}} + \frac{\partial z}{\partial y} \left(\frac{\partial F_z}{\partial x} + \frac{\partial F_z}{\partial z} \frac{\partial z}{\partial x} \right)$$



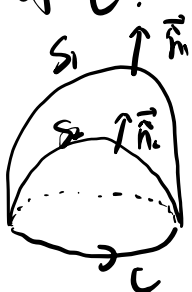
$$\oint_C \vec{F} \cdot d\vec{r} = \iint_R \frac{\partial^2}{\partial x^2} \left(\frac{\partial F_y}{\partial z} - \frac{\partial F_z}{\partial y} \right) + \frac{\partial^2}{\partial y^2} \left(\frac{\partial F_z}{\partial x} - \frac{\partial F_x}{\partial z} \right) + \left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \right) dA.$$

$$\nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix} = \left\langle \frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z}, \frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x}, \frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \right\rangle.$$

$$\begin{aligned} \iint_S (\nabla \times \vec{F}) \cdot \vec{n} \, ds &= \iint_R \left(\left(\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z} \right) \frac{\partial z}{\partial x} - \left(\frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x} \right) \frac{\partial z}{\partial y} + \left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \right) \right) dA. \\ &= \oint_C \vec{F} \cdot d\vec{r} \quad \square \end{aligned}$$

Remarks.

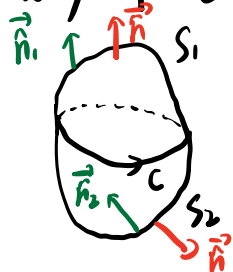
1. A closed curve C is the boundary of two different orientable surface S_1 and S_2 , which both have an orientation consistent with that of C .



$$\iint_{S_1} (\nabla \times \vec{F}) \cdot \vec{n}_1 \, ds = \iint_{S_2} (\nabla \times \vec{F}) \cdot \vec{n}_2 \, ds = \oint_C \vec{F} \cdot d\vec{r}.$$

2. S is a closed surface consisting of S_1 and S_2 with a common

boundary of C ✓



$$S = S_1 \cup S_2.$$

$$\begin{aligned} &\iint_S (\nabla \times \vec{F}) \cdot \vec{n} \, ds \\ &= \iint_{S_1} (\nabla \times \vec{F}) \cdot \vec{n}_1 \, ds - \iint_{S_2} (\nabla \times \vec{F}) \cdot \vec{n}_2 \, ds. \\ &= 0 \end{aligned}$$

3. Conservative vector field $\vec{F} = \nabla f$

Curl test $\nabla \times \vec{F} = 0$.

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot \vec{n} dS = 0$$